NTU ARCHIVE

A Contribution-Based Platform for NTU Past Exams, Homework, and Project Resources

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Live Demo	: https://ntuarchive.vercel.app

TARGET CUSTOMER

1. **Primary Segment**

   NTU students across all departments who need access to past exams, homework, and course materials to prepare for coursework.

2. **Current Status Quo**

   a. Asking professors or TAs directly: inconsistent, many refuse

   b. Asking assigned student mentors (直屬學長姐): depends entirely on having a mentor willing to share, and only covers their specific courses

   c. Scattered PTT/Dcard posts: unstructured, not searchable by course, no quality verification, content disappears into forum history

   d. Word-of-mouth via LINE/Facebook class groups: fragmented per cohort, lost once groups go inactive after the course ends

   e. Anecdotally: international and exchange students may face an added barrier, since most existing platforms and informal channels (Dcard, PTT, class LINE groups) operate primarily in Chinese

3. **Why Better Than Status Quo**

   Past exams today are passed through personal connections. If a student does not know the right person, they do not get the material. This system replaces that with a shared, searchable archive open to every student, regardless of their social network

DEMAND EVIDENCE AND WILLINGNESS TO PAY

1. **Methodology**

   Two independent forms of evidence were collected. First, qualitative and quantitative evidence was gathered from three social platforms — Dcard, PTT, and Threads — by searching for posts related to past exam access (考古題) within NTU specific boards and hashtags. 15 posts were logged with view counts, comment counts, and direct links (full dataset available in the project repository). Second, a short survey was distributed to NTU students using Google Forms, collecting 22 responses across a range of departments (CSIE, chemical engineering, civil engineering, life science, economics, philosophy, and others), asking about past-exam usage habits, perceived value, difficulty of access, and willingness to pay.

2. **Key Findings** (see attachments)

   Forum Evidence

   a. Demand exists. Dcard's own topic page for 考古題 (past exams) shows 4,095 posts and 366 followers under that single tag within the NTU community. This indicates that this is a recurring need across cohorts and years.
   b. High engagement volume on individual posts. A Thread's complaint about exam questions almost entirely matching past exams, reached 4,000 views and 195 comments. A Dcard post titled "大學沒朋友沒考古 題" reached 398 views and 251 comments. Comment counts exceeding view counts indicates high resonance and active discussion.
   c. Past exams function as a competitive necessity. Multiple posts describe direct grade impact. A Threads user reported 11 of 13 exam questions matched past exams exactly, while a Dcard post described professors basing over 70% of exam content on recycled past questions, passed down informally year to year by upperclassmen (直屬學長姐) without faculty awareness.
   d. Access is gated by social capital. This is the clearest evidence supporting the target customer definition. A freshman directly asked the community what channels exist to obtain past exams "besides asking professors or assigned mentors." Another post described emotional distress from exclusion. A student who did not receive past exams from their assigned mentor described crying alone after an exam upon realizing classmates with access had scored higher.

e. Existing informal channels are recognized as insufficient. A Dcard post asked whether it's possible to survive at NTU without access to past exams at all, while a separate post questioned the fairness of the system entirely, noting professors are often unaware their exams are being recycled.

Survey Evidence

a. Past-exam usage. 19 of 22 survey respondents (86%) reported having studied from past exams, with only 3 respondents (concentrated in foreign-language and literature departments) reporting no usage. This shows a pattern consistent with the forum finding that demand is strongest in technically-oriented departments but present broadly across the university.
b. Past exams directly improve academic outcomes. Among respondents who use past exams, the average self-reported rating for "my grade improves after studying from past papers" was 4.63 out of 5, the highest-rated item in the survey, directly supporting the claim that past-exam access has measurable academic value.
c. Difficult access. The average rating for "it's hard to get past papers" was 4.42 out of 5, proving the forum evidence that the current access system, which is primarily through seniors (15 of 19 respondents) and professors voluntarily distributing past material (11 of 19), is inconsistent and effortful even for students who do obtain materials.
d. Willingness to pay is high and quantifiable. The average self-reported willingness to pay was 4.37 out of 5, with 74% of respondents (14 of 19) rating their willingness to pay at 4 or higher. When asked to specify an amount, responses clustered as follows:

Price Range	**Respondents**
50-100 NTD	5
100-200 NTD	7
200-300 NTD	6
300+ NTD	1

3. **Willingness to Pay**

   Forum evidence shows a high non-monetary cost already being paid: active management of inconsistent social relationships (直屬, roommates, classmates) specifically to obtain this material, with measurable academic and emotional cost when that network fails. Survey evidence converts this into a quantifiable monetary figure: a median willingness to pay in the 100–200 NTD range. The contribution-gate model (upload-to-unlock) converts the existing informal "social cost" into a measurable platform action at no monetary cost, while the optional premium tier monetizes the segment of students willing to pay directly to skip that labor, with 100–200 NTD as a reasonable starting price point.

SYSTEM DESIGN

1. **Data Sources**

   The system's data is entirely user-generated. There are no external APIs or scraped data feeds in the live product. Students contribute past exams, homework, and project files directly through an upload form. Each upload captures structured metadata (course code, department, semester, resource type) alongside the file itself, so the catalogue is searchable without relying on manual curation.

2. **Storage and Processing**

   The system uses Supabase, a hosted backend-as-a-service built on PostgreSQL, for three reasons: it provides a relational database, file storage, and authentication in one platform, it requires no server code to be written or maintained, and it is free at the scale needed for this project.

   a. Database (PostgreSQL): Two core tables, **resources** and **contributions**, plus a **profiles table** added later for the merit system.

      i. Resources stores metadata for each uploaded file (title, course code, department, semester, type, file URL, uploader ID).

      ii. Contributions logs every upload event per user, which powers the contribution-gate logic.

      iii. Profiles track per-user nickname, merit balance, and premium status.

   b. File storage: PDFs are stored in a single Supabase Storage bucket (exam-files), separate from the database. The database stores only a reference URL to each file, not the file itself.. A single bucket is used for all resource types (exam, homework, project) rather than one bucket per type. The resource type is a query stored as a database column to keep the storage layer simple and avoids duplicating upload logic and security policies across multiple buckets.

   c. Access control (Row Level Security): Every table enforces PostgreSQL Row Level Security policies rather than relying on the frontend to restrict access. For example, a user can only delete a resource they uploaded, and merit changing operations run through dedicated database functions (*award_upload_merit*, *spend_preview_merit*) rather than direct table writes, so a user cannot manipulate their own merit balance by bypassing the frontend.

d. Processing logic: Search and filtering (by department, type, course code) happen client-side over the fetched resource list, which is appropriate at this scale. The merit system is the system's main piece of processing logic: uploading awards merits, previewing another student's file deducts merits from the viewer and credits the uploader, and these transactions are executed atomically inside PostgreSQL functions to prevent inconsistent states

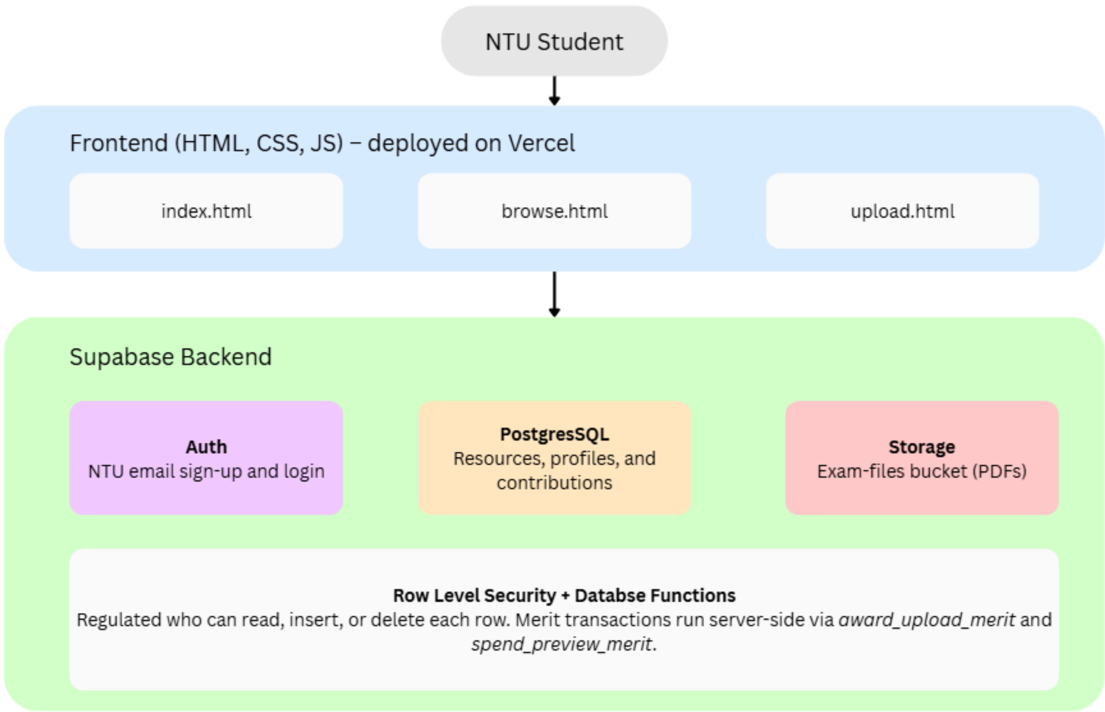
3. Delivery

The product is delivered as a web application (HTML, CSS, JavaScript), chosen over an app for a simpler access and because file upload/download workflows are better suited to a browser. Three pages are at the core flow:

a. Sign in/sign-up page which are restricted to @ntu.edu.tw email addresses
b. Browse page with filtering and an in-browser PDF preview
c. Upload page that writes to both storage and the database.

The site is deployed on Vercel and publicly accessible at https://ntuarchive.vercel.app.

4. Architecture Diagram



GO-TO-MARKET DIFFICULTIES

1. **Start Problem**

   A new platform asking students to upload past exams has no track record nor worthy content. Early adoption likely depends on seeding the catalogue with a critical mass of content before most students see value in joining. Department-by-department rollout, starting with departments with strong demand evidence is a more realistic strategy than a campus-wide launch on day one.

2. **Data Acquisition and Account Abuse**

   During development, testing revealed that Supabase's signup process creates an account record immediately upon request, before the email confirmation step. This means the platform is technically exposed to automated signup spam using fake or pattern-generated @ntu.edu.tw-style addresses, even though such accounts cannot log in or take any action until confirmed. A production deployment would need rate limiting on signups and periodic cleanup of unconfirmed accounts.

3. **Enforcement Gaps in The Merit System**

   The merit-gated preview feature is enforced at the application layer (a database function checks and deducts merits before returning file access), not at the storage layer. A user could, in principle, bypass the merit cost by directly accessing a file's storage URL instead of using the Preview button. To address this, it would require issuing short-lived signed URLs per preview request, which was limited due to time and resources I have, but is a suggestion for next improvement.

4. **Legal and Content Concerns**

   Past exams are technically the intellectual property of the instructor who wrote them, not the student who took the exam. The platform currently relies on voluntary student contribution without a formal rights-clearance process. A production version would need a clear takedown policy and likely direct outreach to departments to establish whether exam-sharing is permitted, discouraged, or actively against course policy. This varies by professor and was not formally tested as part of this project.

5. **Competition**

   Platforms like Docsity and Course Hero already provide past exams from many universities, but they are not designed specifically for NTU, do not focus on Chinese content, and use a standard paywall instead of a contribution-based system. In contrast, this system is built around NTU's course structure and encourages students to contribute materials in exchange for access, creating a community-driven model that general-purpose platforms are unlikely to support for a single university.

6. **Unit Economics**

   At current scale (manual Supabase free tier), the system has no real operating cost. At scale, the main costs would be storage (PDF file size × number of uploads) and database usage, both of which scale with active users rather than fixed infrastructure, making the cost structure naturally aligned with usage-based premium subscription revenue if growth occurs

FUTURE IMPROVEMENT

1. Expand ways to earn merit. Users could gain merit not only by uploading files but also by contributing course reviews or other useful content. This would allow students without past exams to participate and encourage collaboration with other NTU community platforms.
2. Improve content moderation. Add a reporting system and administrator review tools to detect duplicate, incorrect, or inappropriate uploads and maintain content quality.
3. Introduce reputation features. Allow users to rate uploads or recognize trusted contributors so that high-quality resources become easier to identify.

AI USAGE

Mainly use Claude to:

1. Help write the UI/UX
2. Setting up the user policies
3. Make the architecture diagram
4. Ask questions related to setting up the database